Swiss Institute of Bioinformatics

PR STATS

Metagenomic Data Analysis of Microbial Communities (MGMC01)

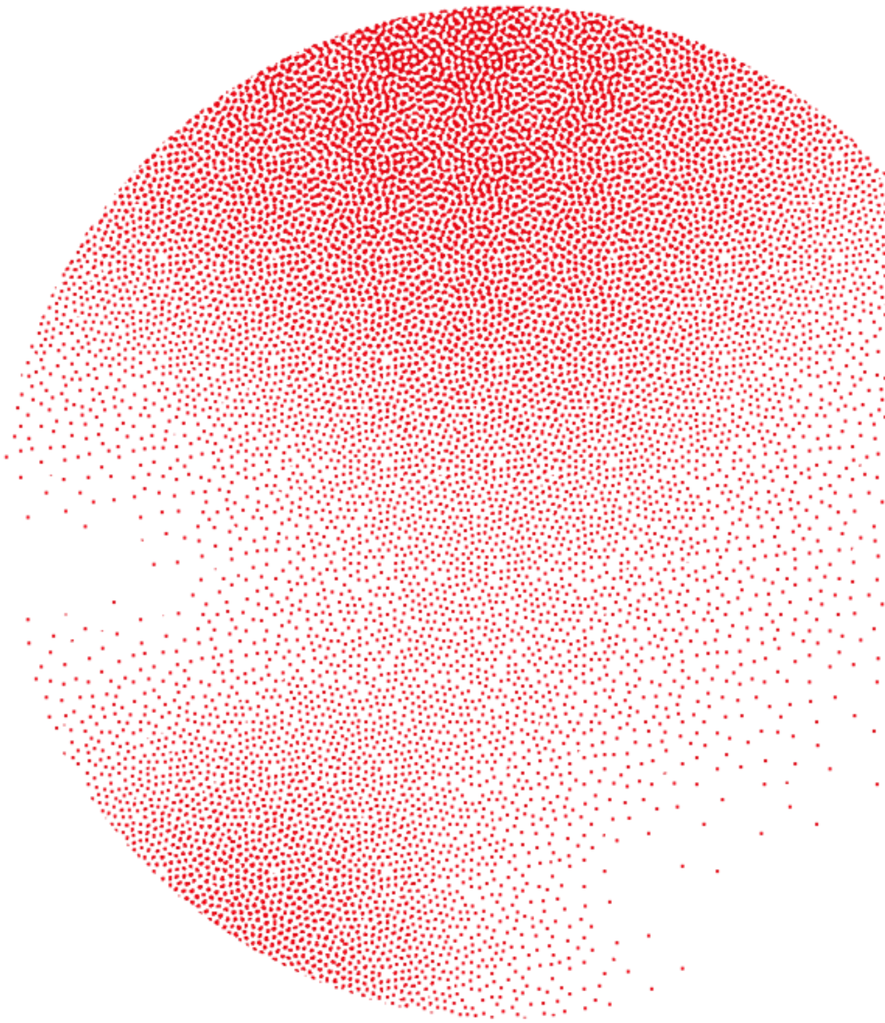
Day 4 - Session 1

PhD Jeferyd Yepes-García
University of Fribourg
October 29th, 2026

elixir SWITZERLAND

This presentation

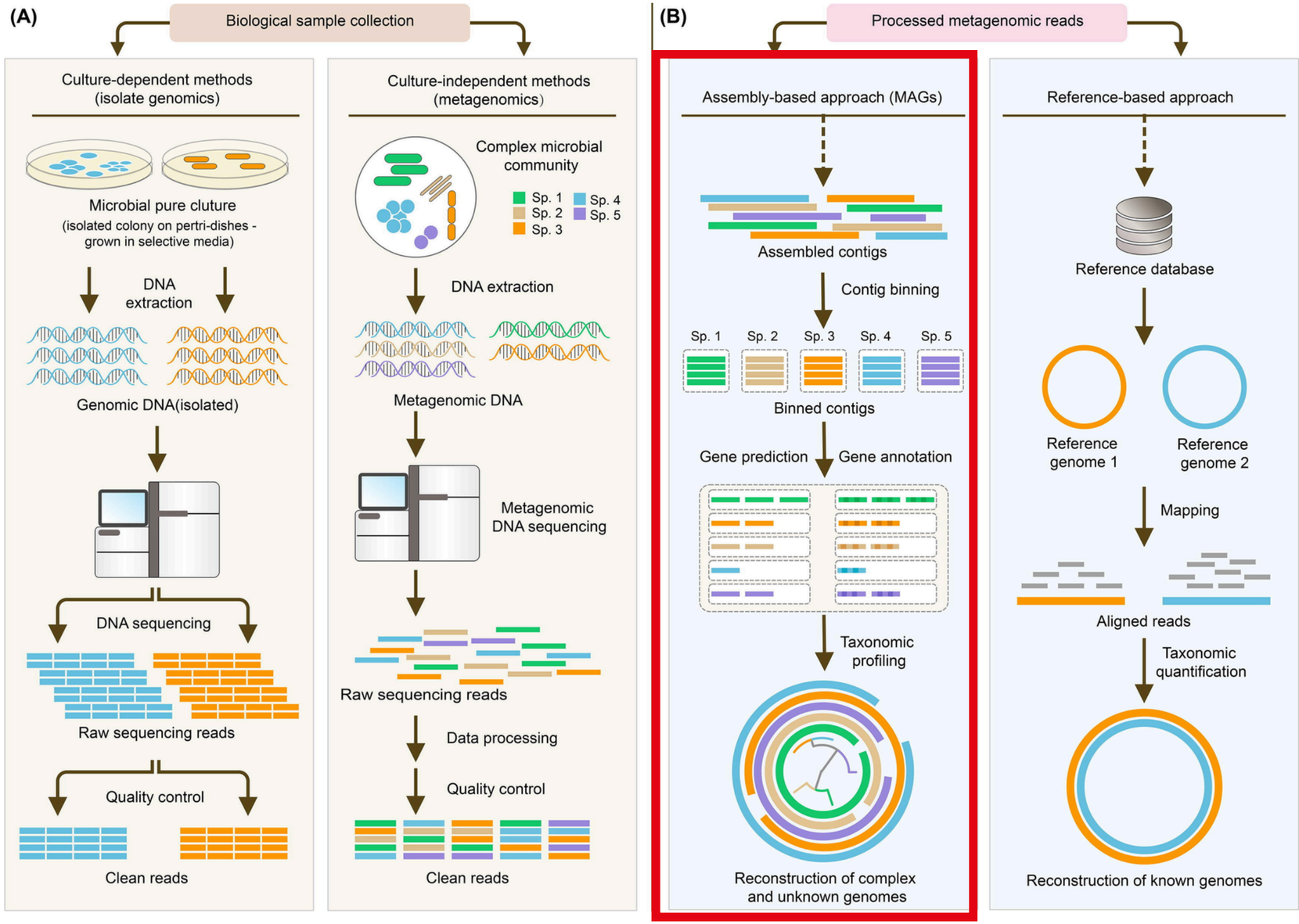
- Learning outcomes
- Metagenome assembly
- Binning
- Bin refinement
- Quality assessment
- Pipelines
- Summary



Learning outcomes

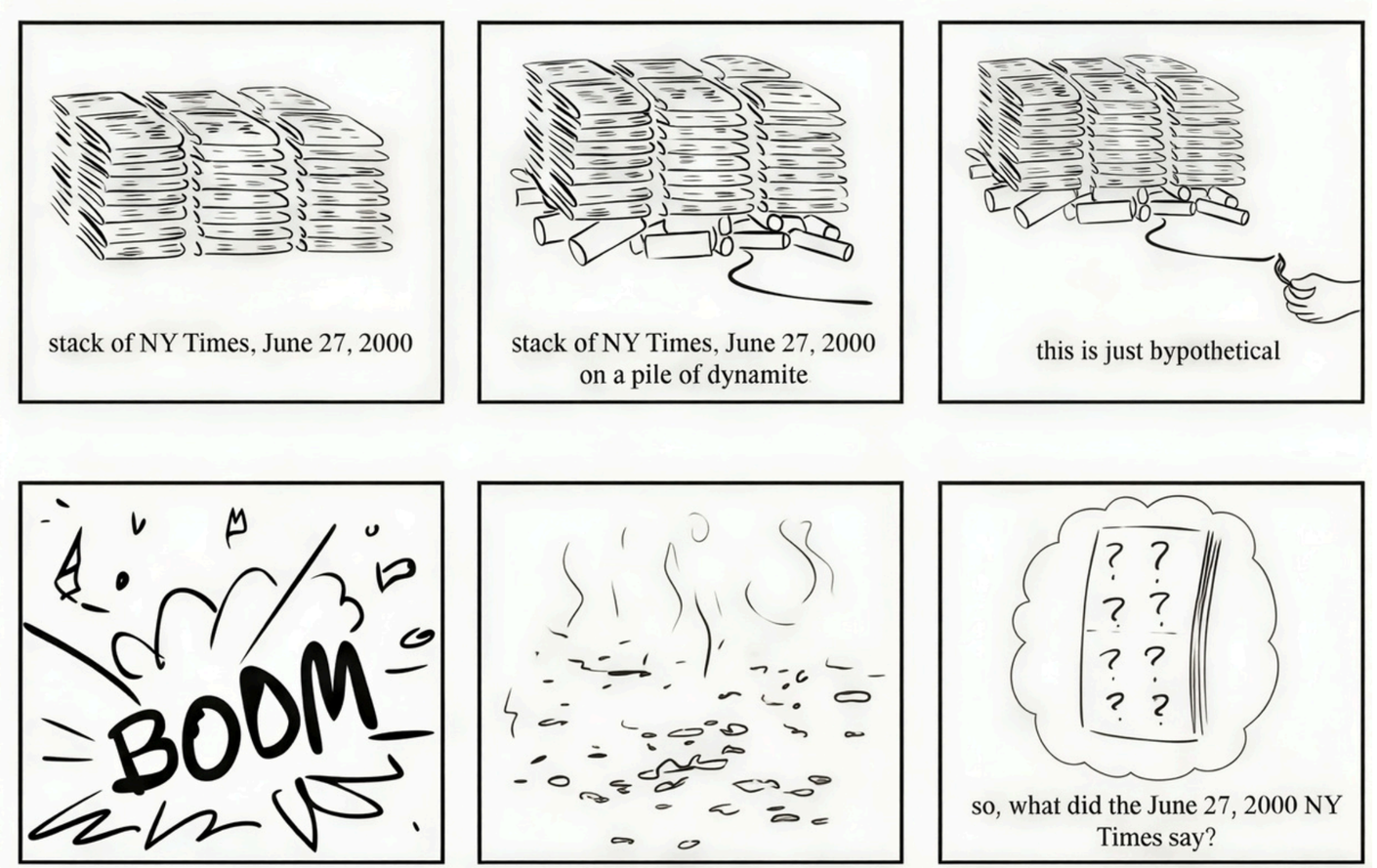
- Explain the principles of *de novo* metagenomic assembly, as well as binning strategies for recovering individual genomes from assemblies.
- Discuss the challenges that community complexity, strain microdiversity and uneven coverage pose for assembly.
- Understand different methodologies to evaluate genome quality.
- Recognize the importance of using orchestrated pipelines to assure reproducibility across studies.

How to study a microbiome/metagenome?



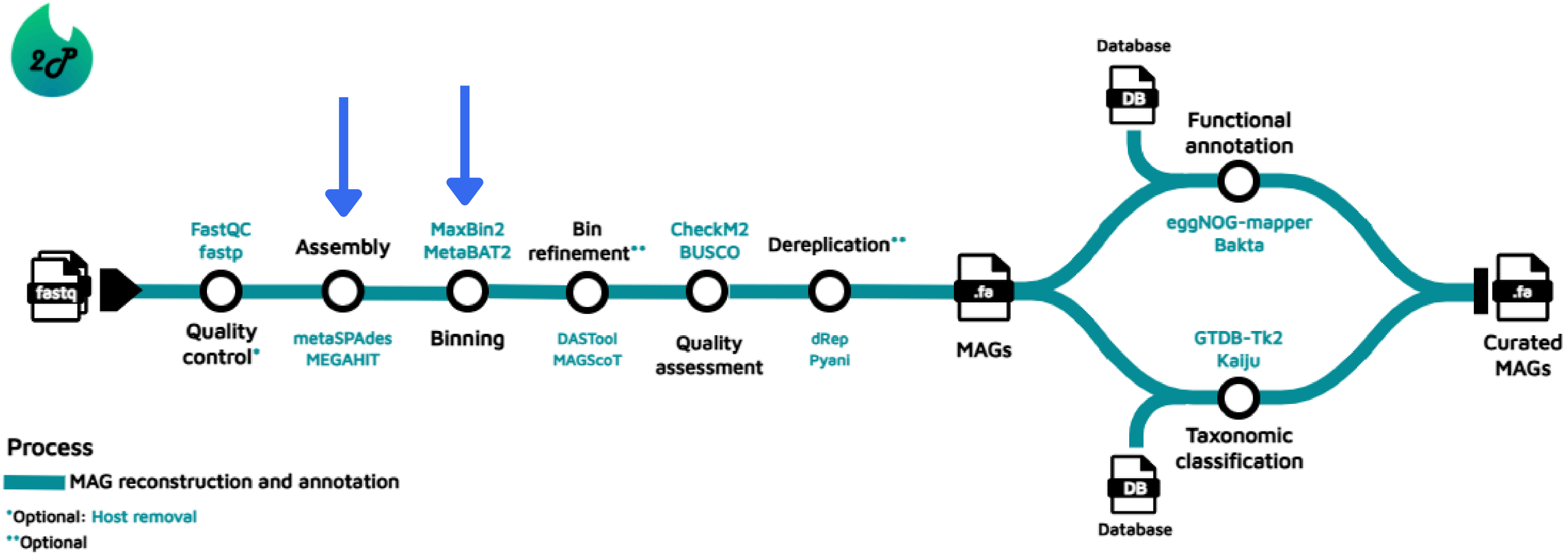
Metagenome assembly

How could we use the tiny snippets of newspaper to figure out what the news was on June 27, 2000?



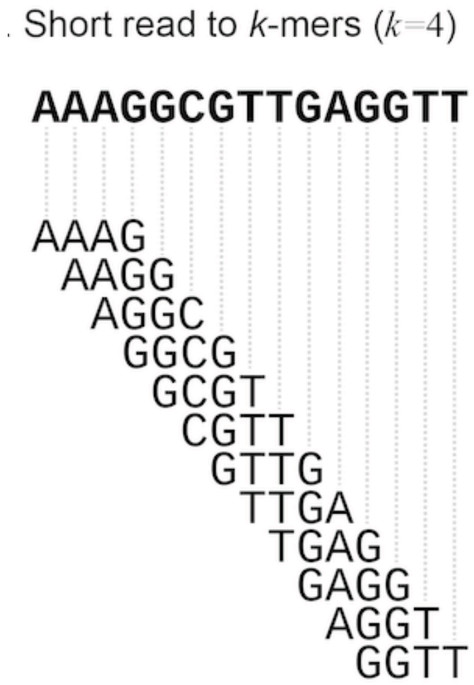
Phillip Compeau & Pavel Pevzner (2014)

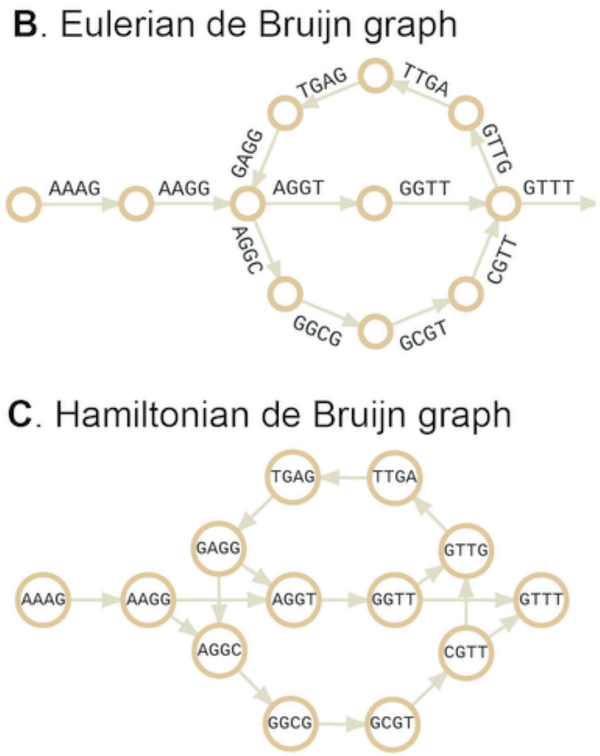
Common MAG-oriented bioinformatics workflow

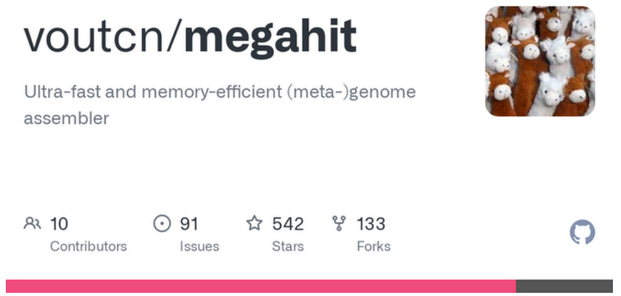


Assembly

Reconstruction of longer DNA sequences (contigs/scaffolds) by merging overlapping short sequencing reads.



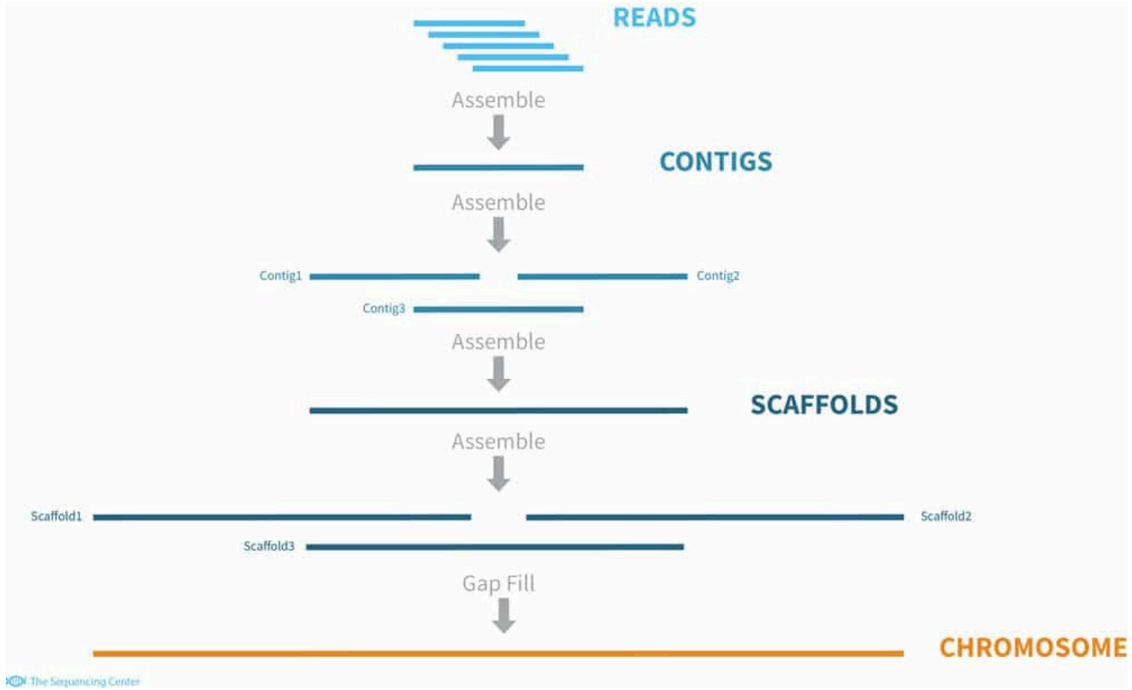


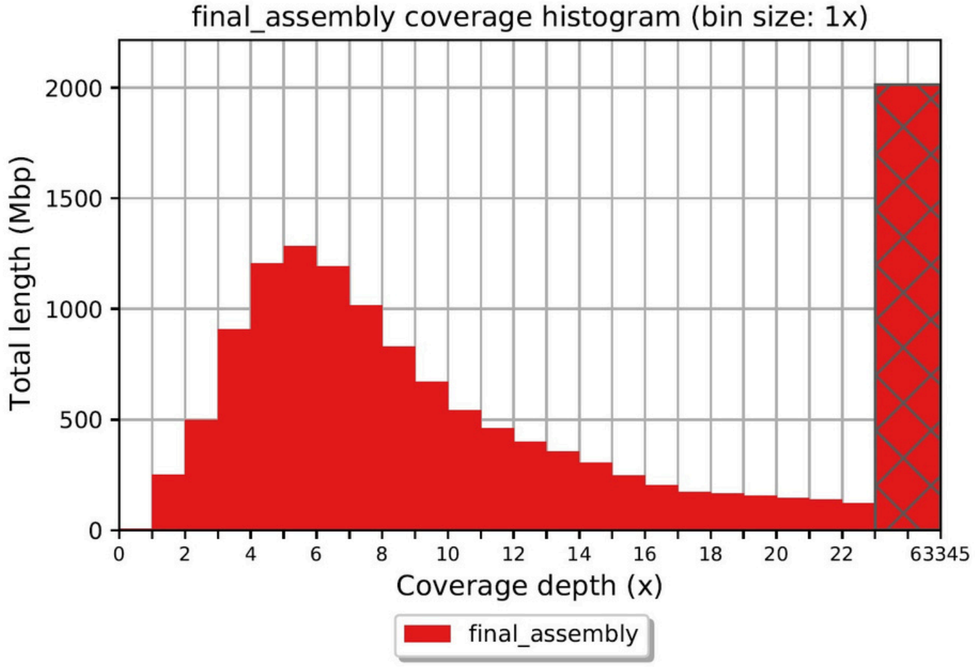




Assembly

Reconstruction of longer DNA sequences (contigs/scaffolds) by merging overlapping short sequencing reads.

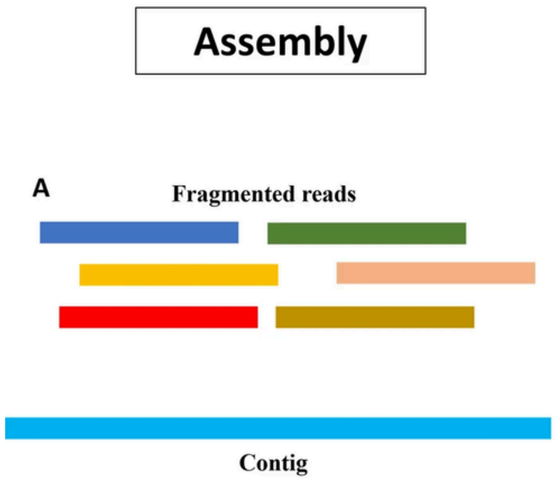


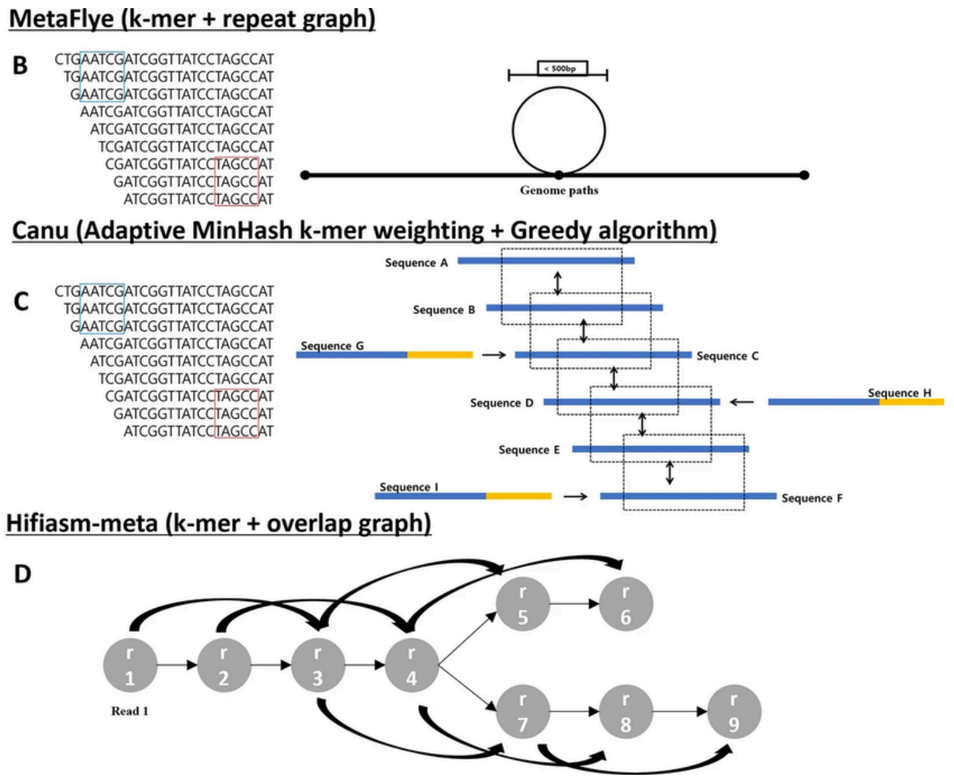


Reality in complex environments

Assembly

Long reads: Usually, the methods include repeat graphs, overlap-layout-consensus and overlap graphs..





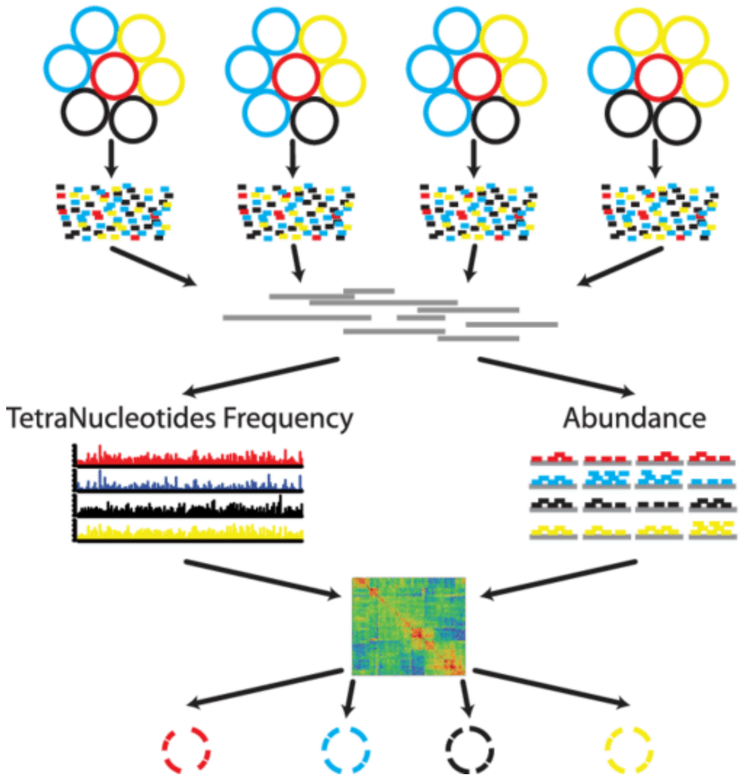
Other assemblers:

- myloasm
- metaMDBG
- Strainy

Kim et al. (2024)

Binning

Grouping assembled contigs into clusters (bins) that represent putative genomes.



Preprocessing

1. Samples from multiple sites or times
2. Metagenome libraries
3. Initial de-novo assembly using the combined library

MetaBAT

4. Calculate TNF for each contig
5. Calculate Abundance per library for each contig
6. Calculate the pairwise distance matrix using pre-trained probabilistic models
7. Forming genome bins iteratively

BigDataBiology/
SemiBin

SemiBin: metagenomics binning with self-supervised deep learning

5 Contributors · 12 Issues · 1 Discussion · 102 Stars · 11 Forks

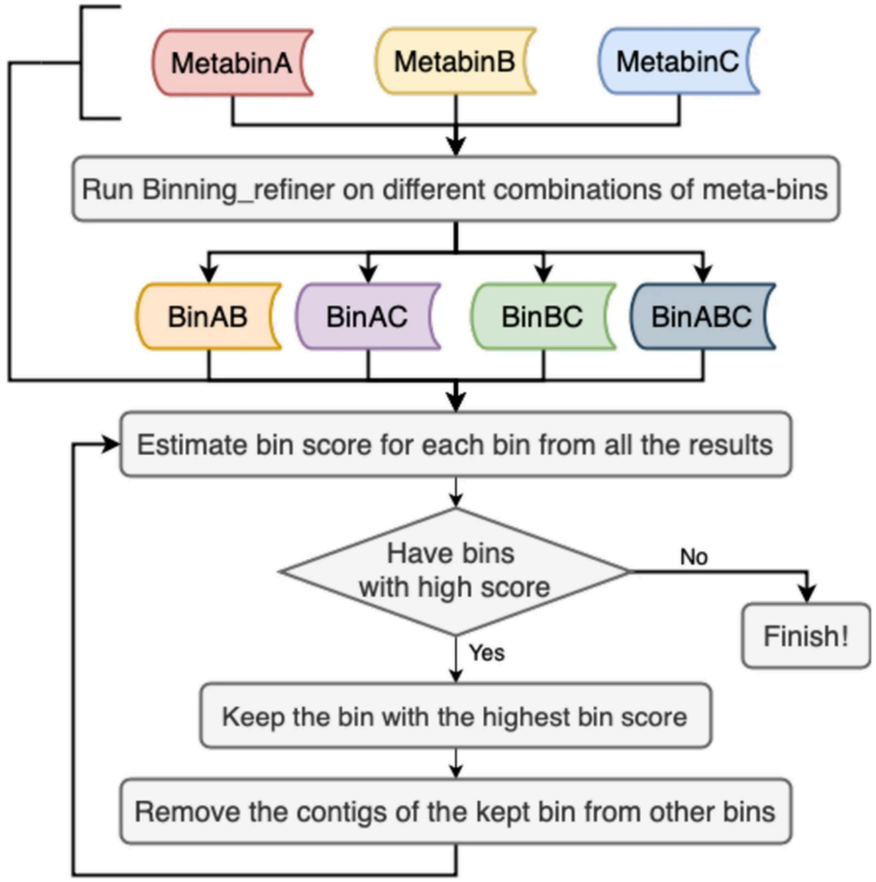
ziyewang/
MetaBinner

2 Contributors · 8 Issues · 39 Stars · 5 Forks

Kang et al. (2015)

Bin refinement

Combining and improving the output of multiple metagenomic binning tools to generate higher-quality MAG.



Tools:

- DASTool
- MAGScot
- MetaWRAP (refinement)
- MAGmax
- Binette
- MetaBinner

Wang et al. (2023)

Quality assessment

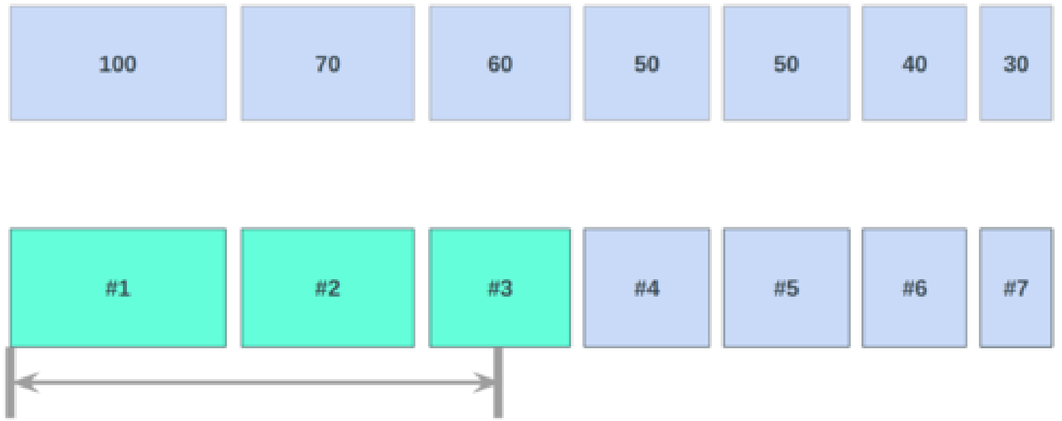
QUAST: Evaluates assembly quality using metrics like N50, contig count, and genome size.

Aligned to "BCep_ref" | 8 605 945 bp | 4 fragments | 66.61% G+C
7705 genomic features

Worst Median Best ☑ Show heatmap

Genome statistics	spades_default	spades_kmers_careful	megahit_default	megahit_min_count_3
Genome fraction (%)	98.14	98.421	98.6	98.603
Duplication ratio	1	1	1.001	1.001
# genomic features	7539 + 75 part	7563 + 62 part	7540 + 105 part	7540 + 104 part
Largest alignment	455 950	505 898	350 746	350 746
Total aligned length	8 436 553	8 470 789	8 492 473	8 493 177
NGA50	143 431	198 969	144 083	125 159
LGA50	18	13	20	21
Misassemblies				
# misassemblies	6	6	9	8
Misassembled contigs length	1 469 048	1 719 134	1 200 775	1 050 989
Mismatches				
# mismatches per 100 kbp	3.85	2.72	2.38	2.3
# indels per 100 kbp	0.67	0.45	0.32	0.28
# N's per 100 kbp	0	0	0	0
Statistics without reference				
# contigs	132	91	156	158
Largest contig	754 490	961 949	539 126	539 126
Total length	8 447 218	8 472 540	8 492 975	8 493 797
Total length (>= 1000 bp)	8 447 218	8 472 540	8 492 975	8 493 797
Total length (>= 10000 bp)	8 324 069	8 384 754	8 296 360	8 308 919
Total length (>= 50000 bp)	7 438 644	7 723 080	6 917 725	6 910 273

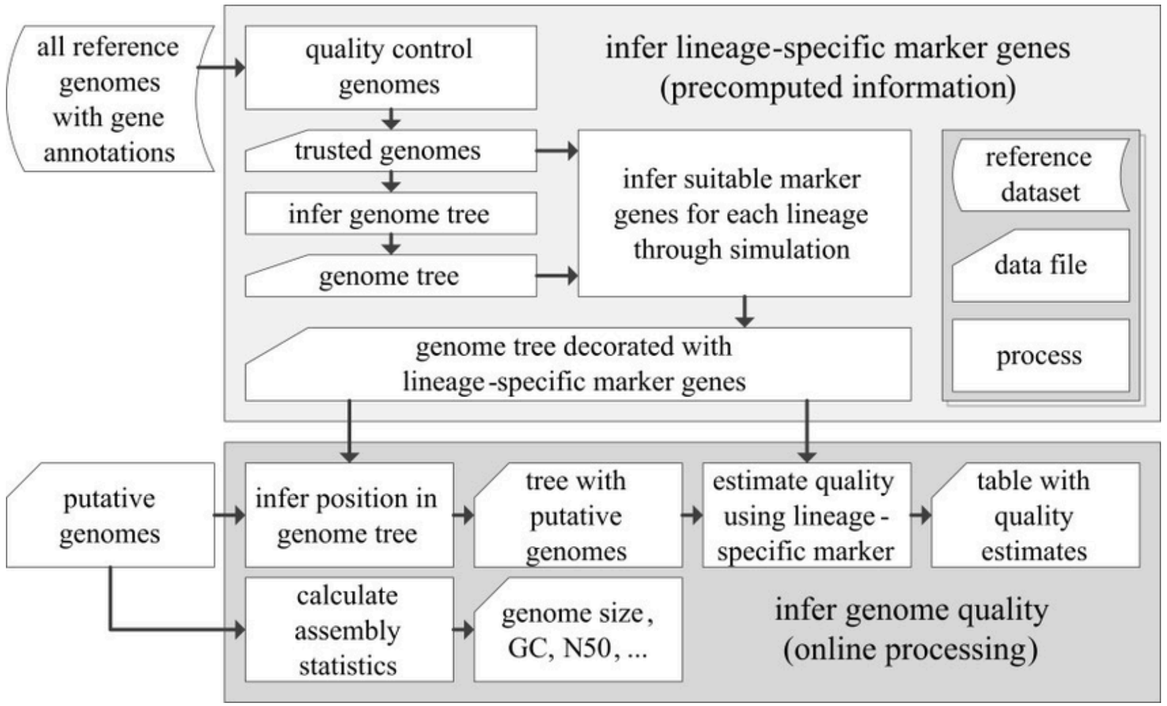
L50 = 3 **N50 = 60**



Mikheenko et al. (2016)

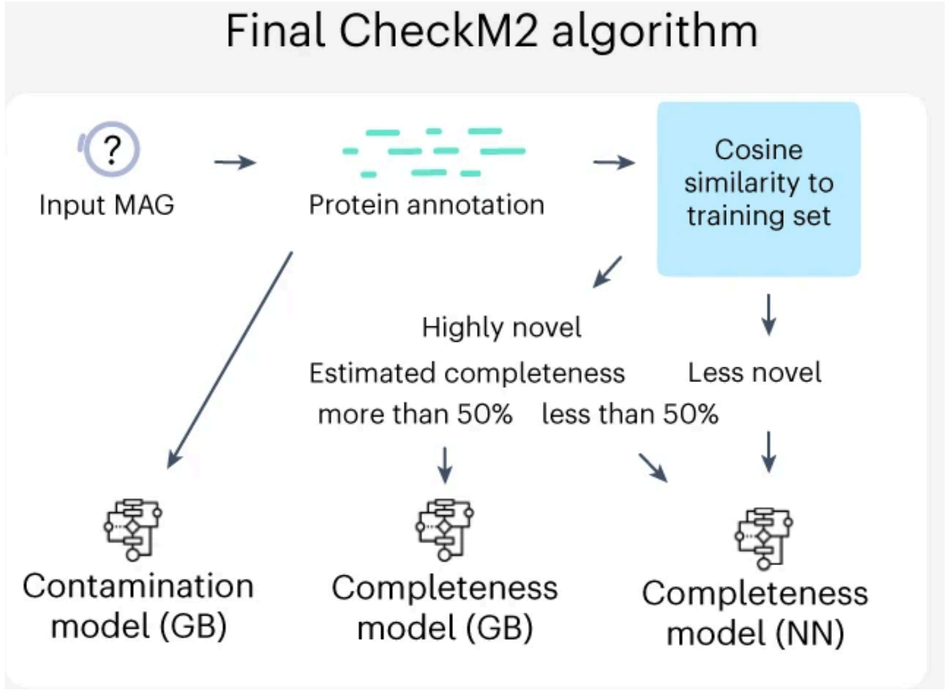
Quality assessment

CheckM: Estimates completeness and contamination of MAGs using lineage-specific marker genes.



Parks et al. (2015)

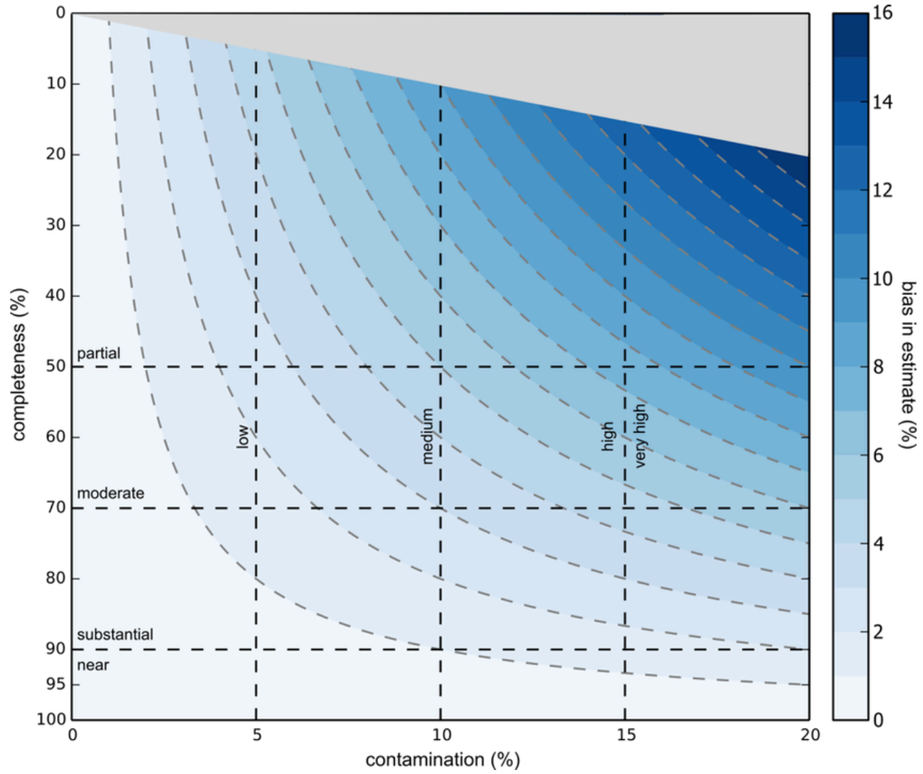
CheckM2: Predicts genome quality of MAGs using machine learning.

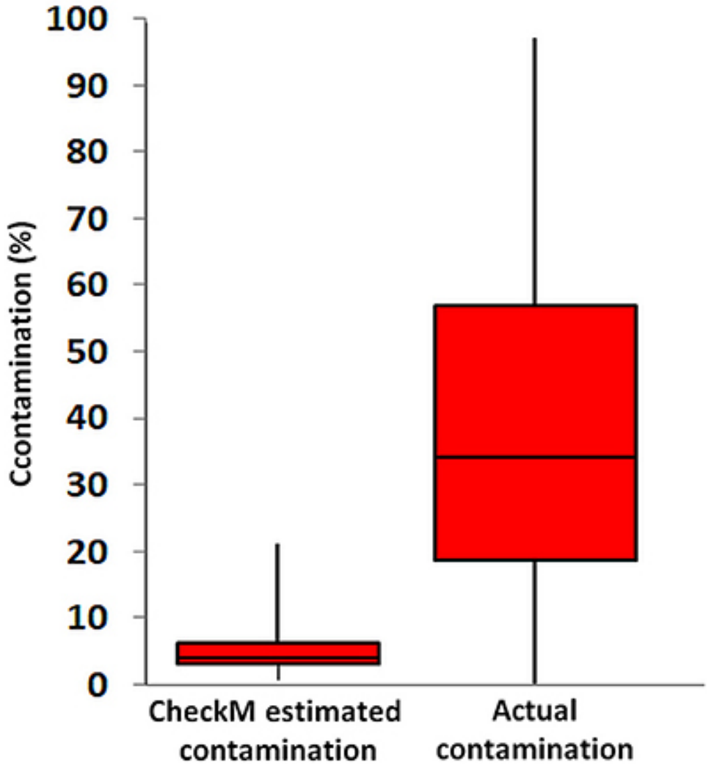


Chklovski et al. (2023)

Quality assessment

Importance of completeness to correctly estimate genomic contamination when using marker gene-based methods.





Quality assessment

MIMAG criteria: Guidelines defining quality tiers (high, medium, low) for MAGs based on completeness, contamination, and other metrics.

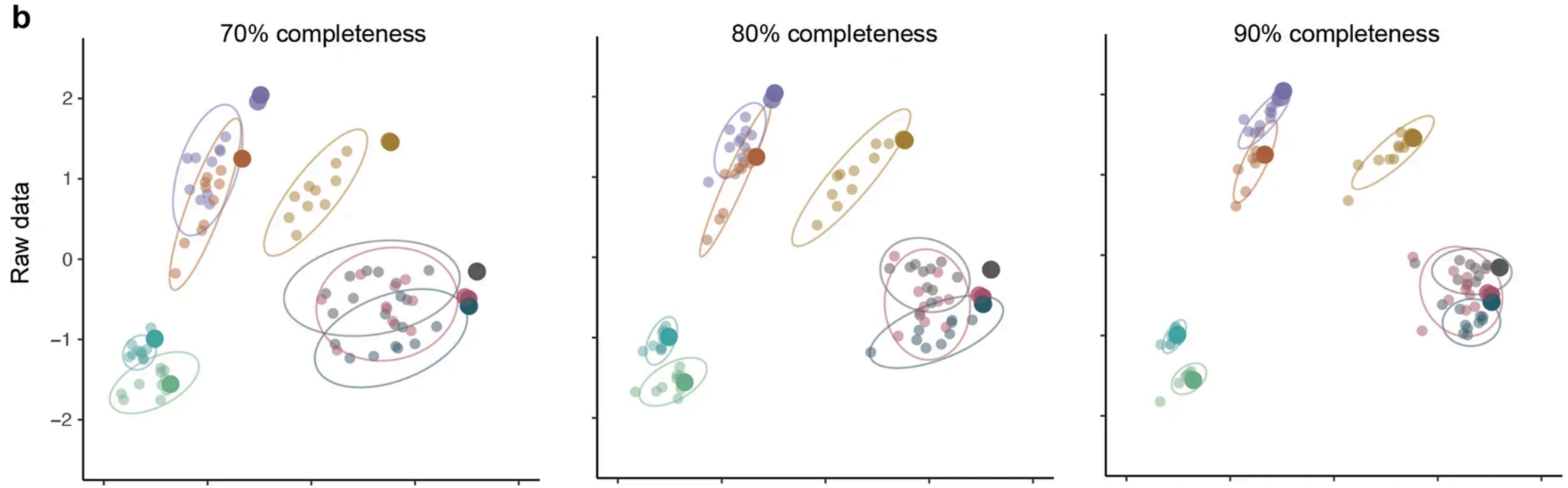
Table 1 Genome reporting standards for SAGs and MAGs

Criterion	Description
Finished (SAG/MAG)	
Assembly quality[a]	Single contiguous sequence without gaps or ambiguities with a consensus error rate equivalent to Q50 or better
High-quality draft (SAG/MAG)	
Assembly quality[a]	Multiple fragments where gaps span repetitive regions. Presence of the 23S, 16S, and 5S rRNA genes and at least 18 tRNAs.
Completion[b]	>90%
Contamination[c]	<5%
Medium-quality draft (SAG/MAG)	
Assembly quality[a]	Many fragments with little to no review of assembly other than reporting of standard assembly statistics.
Completion[b]	≥50%
Contamination[c]	<10%
Low-quality draft (SAG/MAG)	
Assembly quality[a]	Many fragments with little to no review of assembly other than reporting of standard assembly statistics.
Completion[b]	<50%
Contamination[c]	<10%

Bowers et al. (2017)

Quality assessment

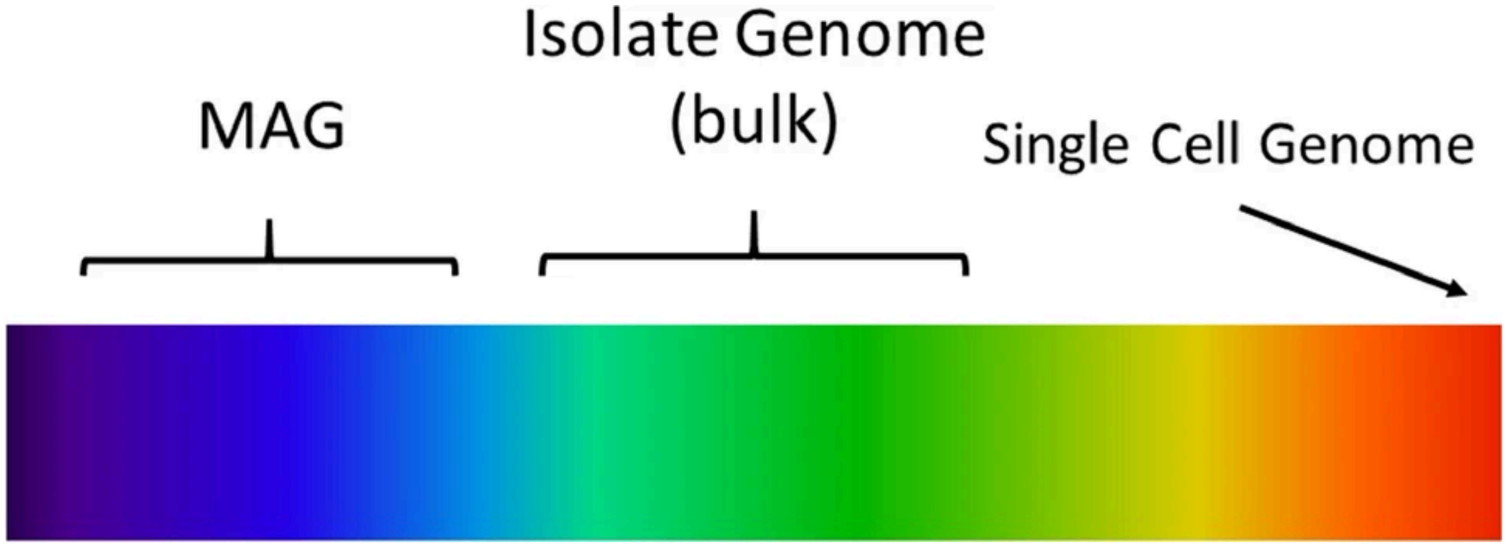
Impact of microbial genome completeness on metagenomic functional inference: Completeness of genomes has a significant impact on the recovered functional signal, spanning all domains of metabolic functions.



Eisenhofer et al. (2023)

Quality assessment

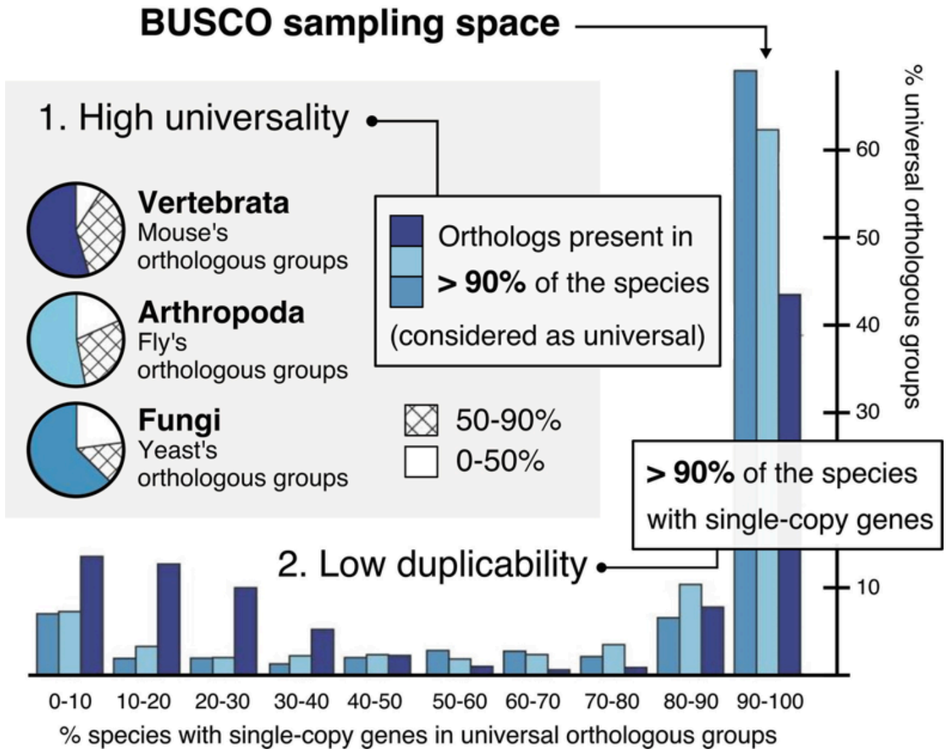
Genome heterogeneity spectrum: MAGs for which a species can be assigned (SMAGs) and MAGs for which this is not possible, because they supposedly are genomes of novel species (HMAGs).

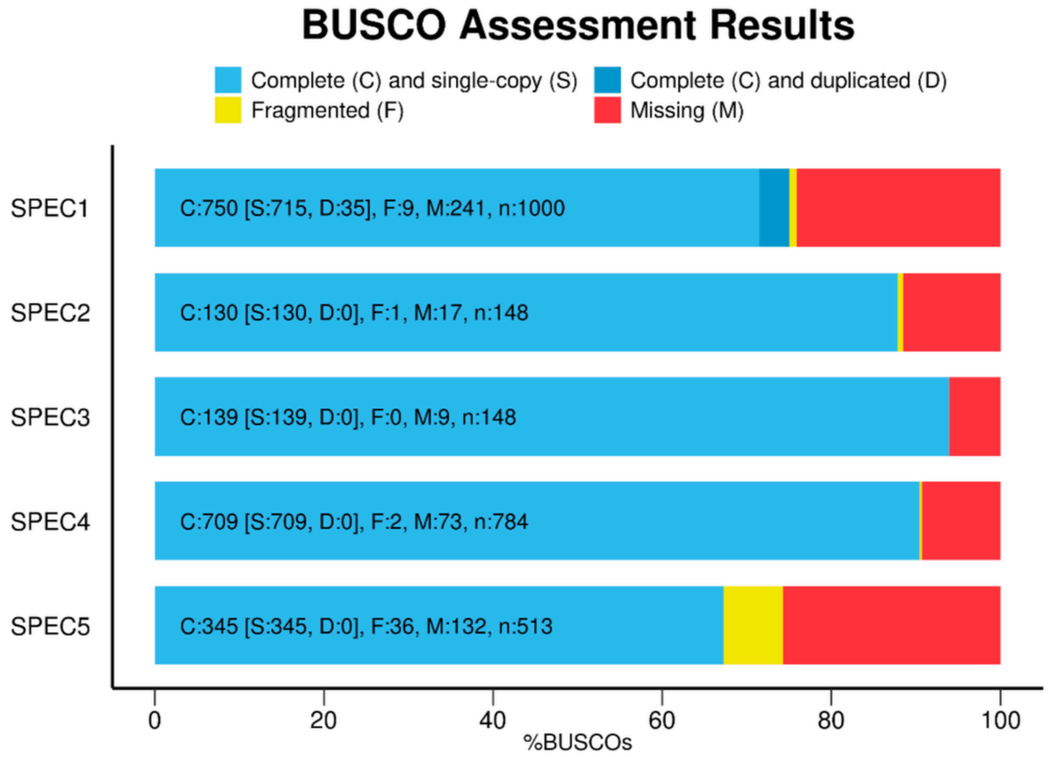


Setubal (2021)

Quality assessment

BUSCO: Assesses genome completeness based on the presence of conserved single-copy orthologous genes.

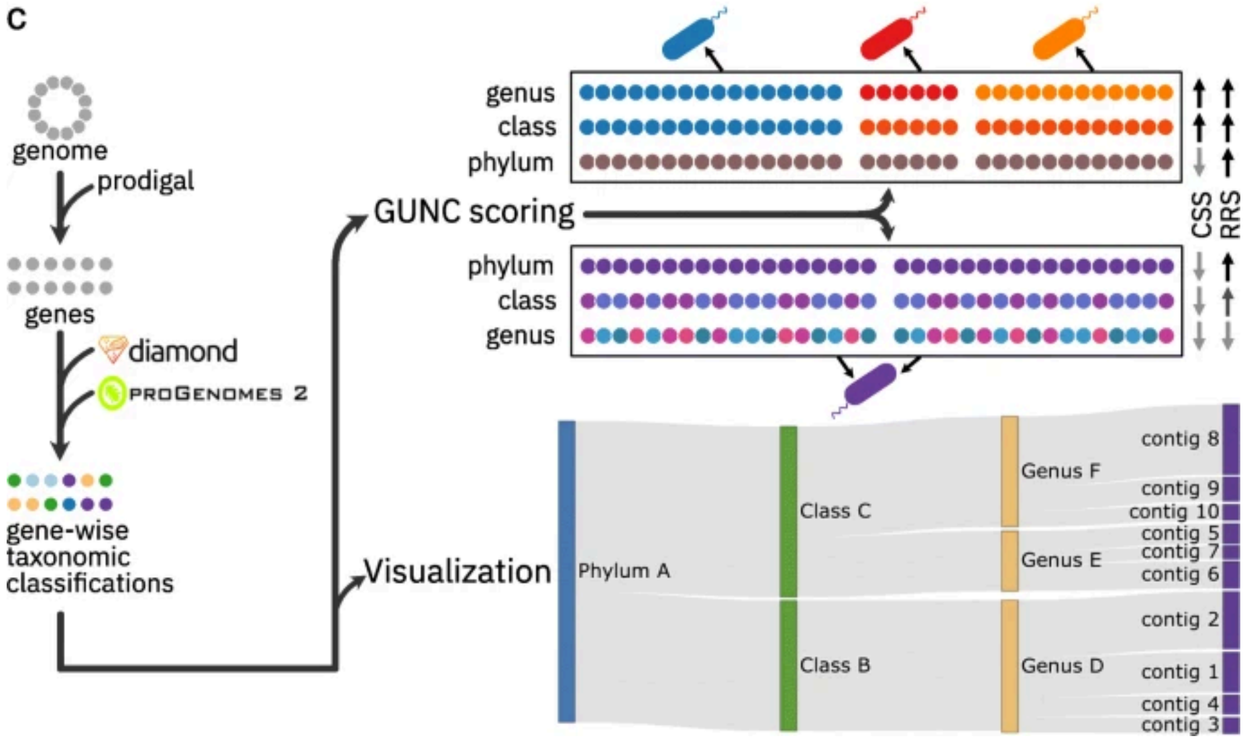




Simao (2015)

Quality assessment

GUNC: Detects chimerism and contamination in genomes by identifying inconsistent taxonomic signals across genes.

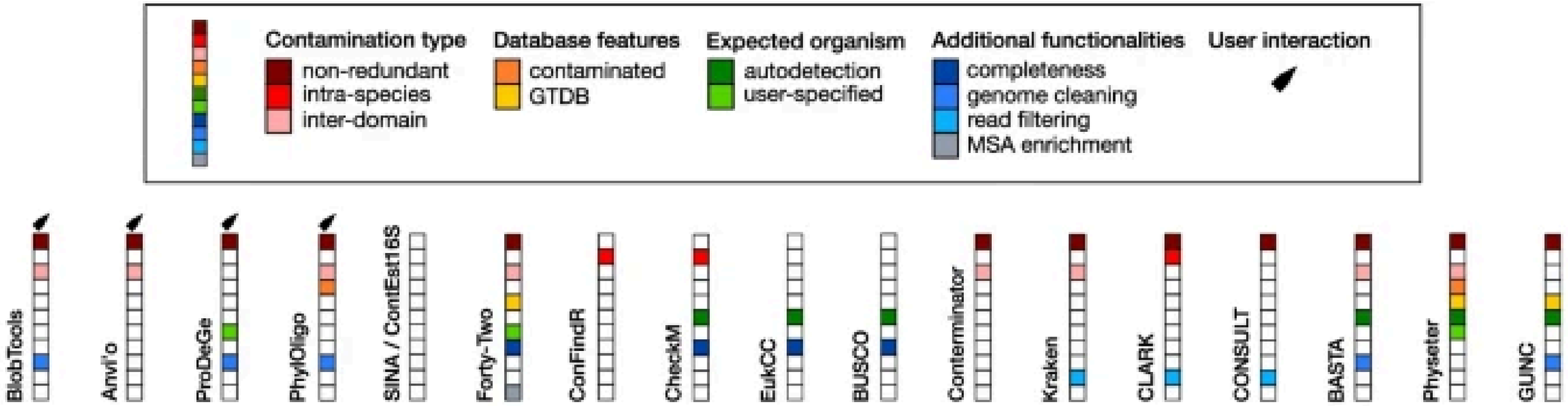


Orakov (2021)

The **Clade Separation Score** (CSS) quantifies the degree to which a genome is a chimeric mixture of distinct lineages following non-random distributions across contigs.

Quality assessment

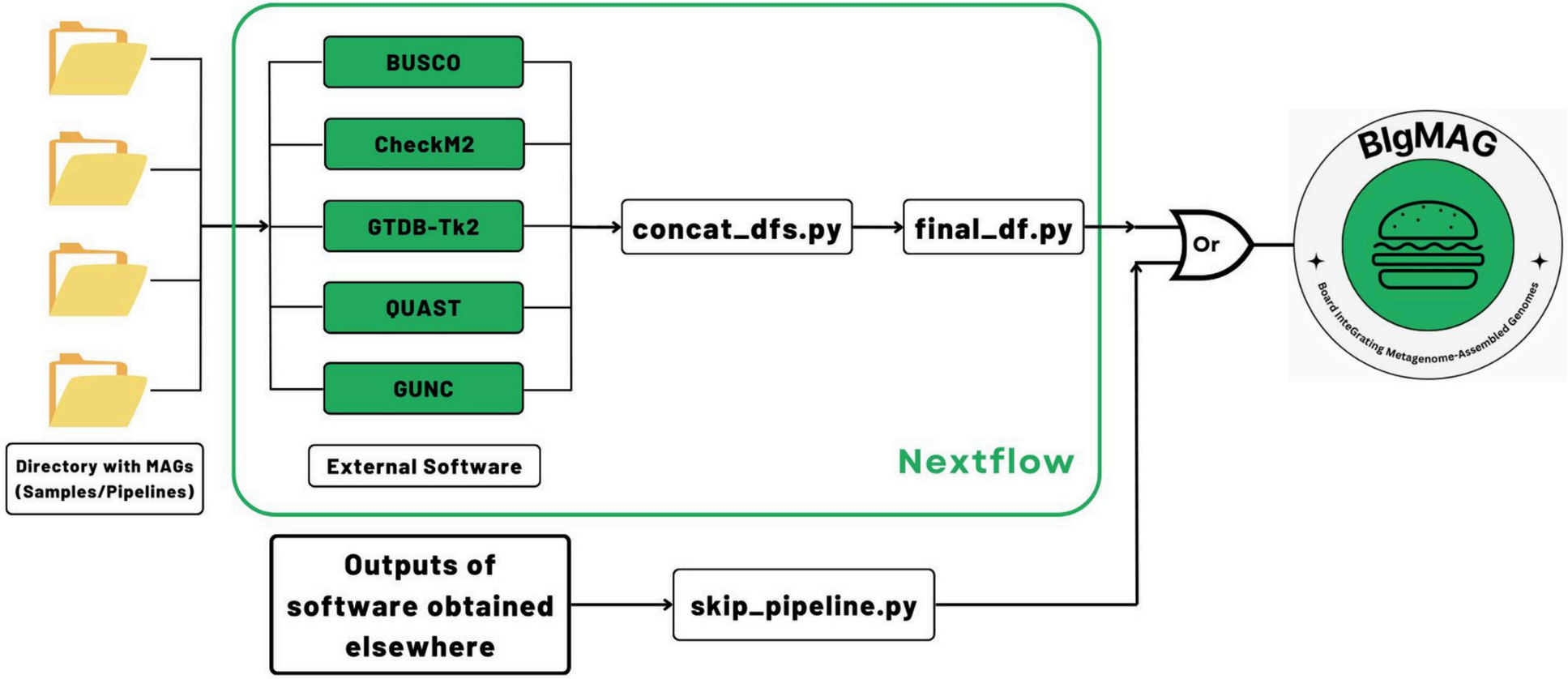
“Contamination detection in genomic data: more is not enough”



Cornet & Baurain (2022)

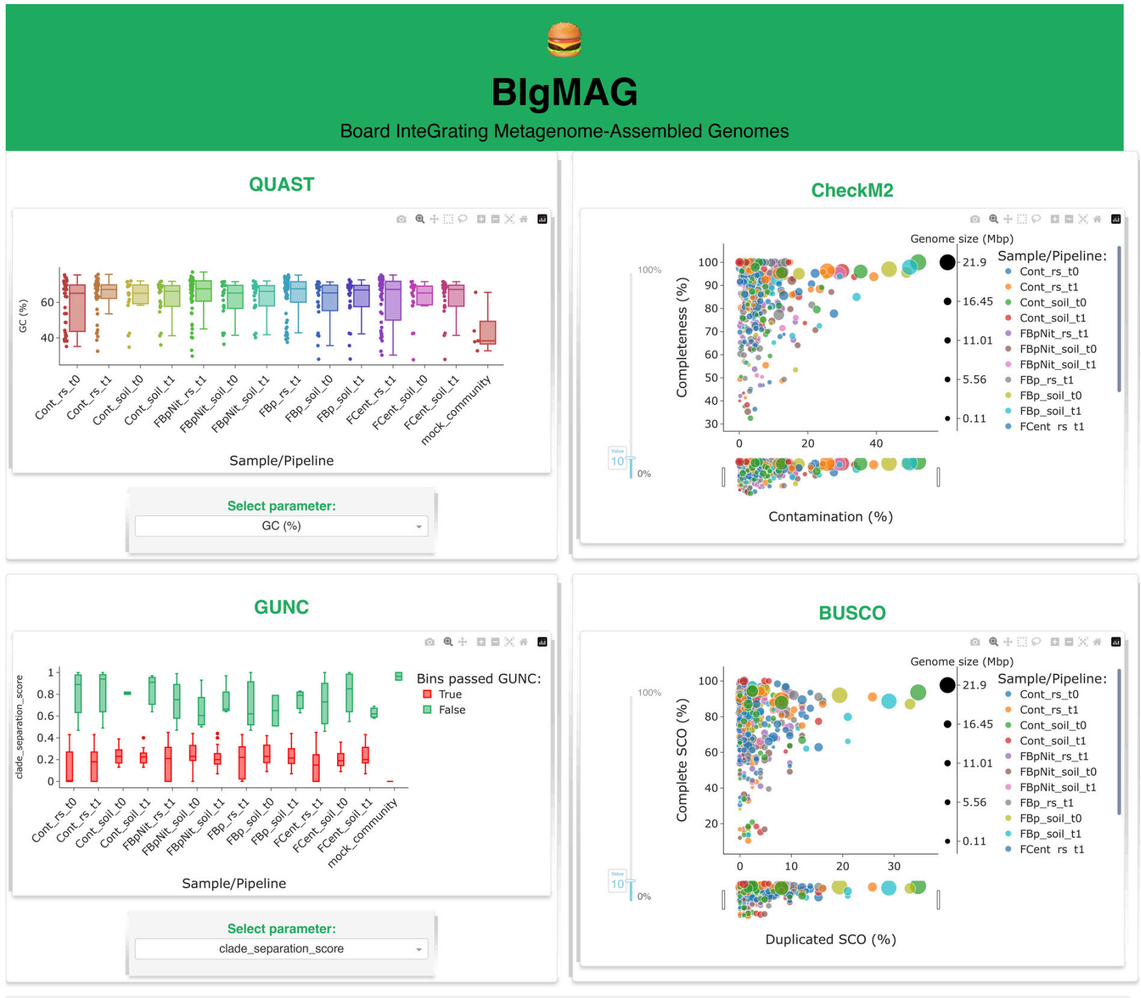
Quality assessment

MAGFlow/BIgMAG: Metagenome quality metrics and taxonomical annotation visualization.



Quality assessment

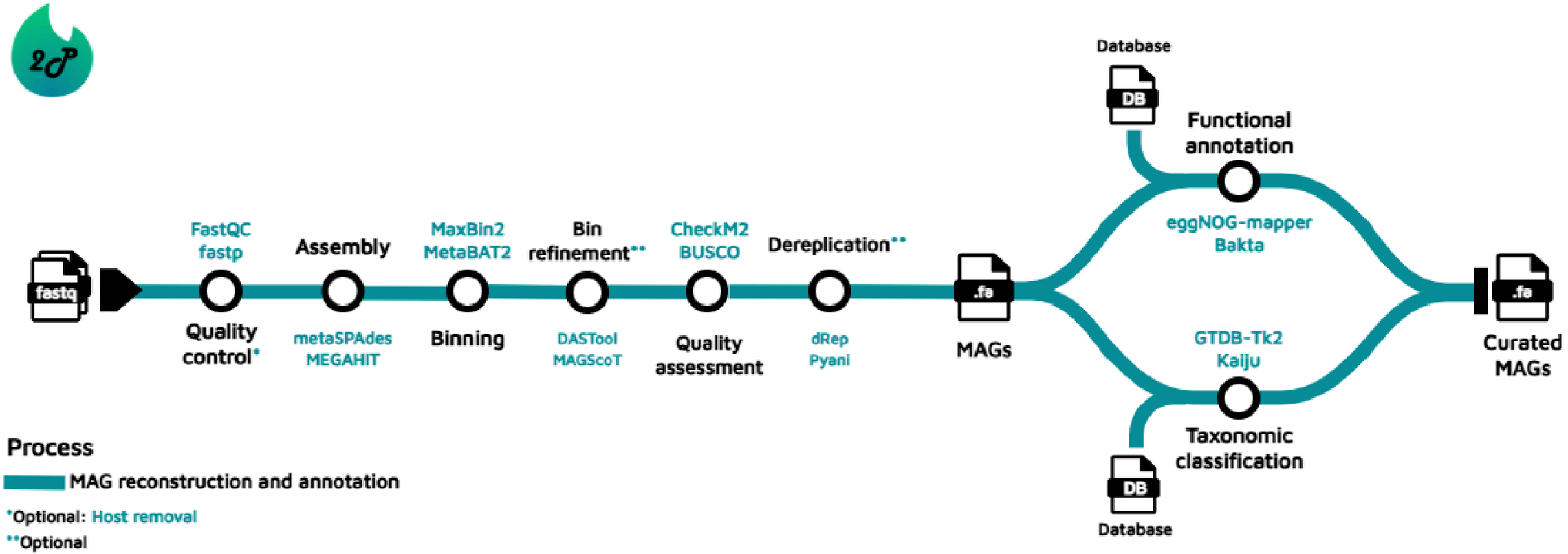
Dashboard:



Alternatives:

- MAGqual
- PANKEGG
- metashot/prok-quality
- CAMI Benchmarking Portal

Common MAG-oriented bioinformatics workflow



Pipeline selection

More than 50 documented pipelines to reconstruct MAGs from environmental samples!



Today's data:

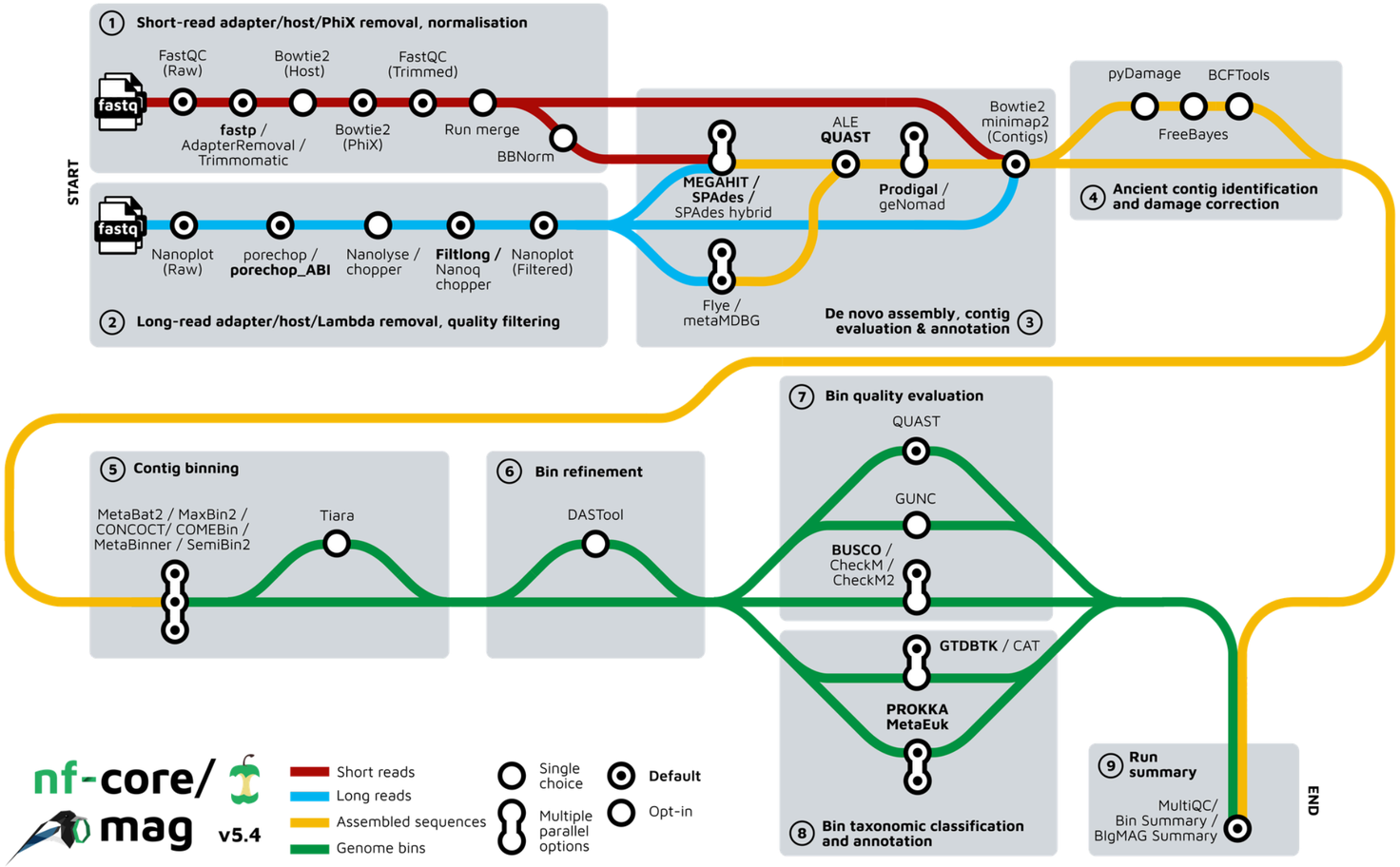
The dataset we will be using for this part of the workshop was collected in **Cuatro Ciénegas (México)**. Cuatro Ciénegas is an oasis in the Mexican desert whose environmental conditions are often linked to the ones present in ancient seas, due to a **higher-than-average content of sulfur and magnesium** but a **lower concentrations of phosphorus and other nutrients** (Okie et al. 2020); the BioProject accession number: PRJEB22811.

Name	Group	Replicate
ERR2143758	Cont	1
ERR2143759	Cont	2
ERR2143760	Cont	3
ERR2143771	Unenriched	1
ERR2143772	Unenriched	2
ERR2143774	Unenriched	3
ERR2143769	Fertilized	1
ERR2143770	Fertilized	2
ERR2143773	Fertilized	3



Pipeline

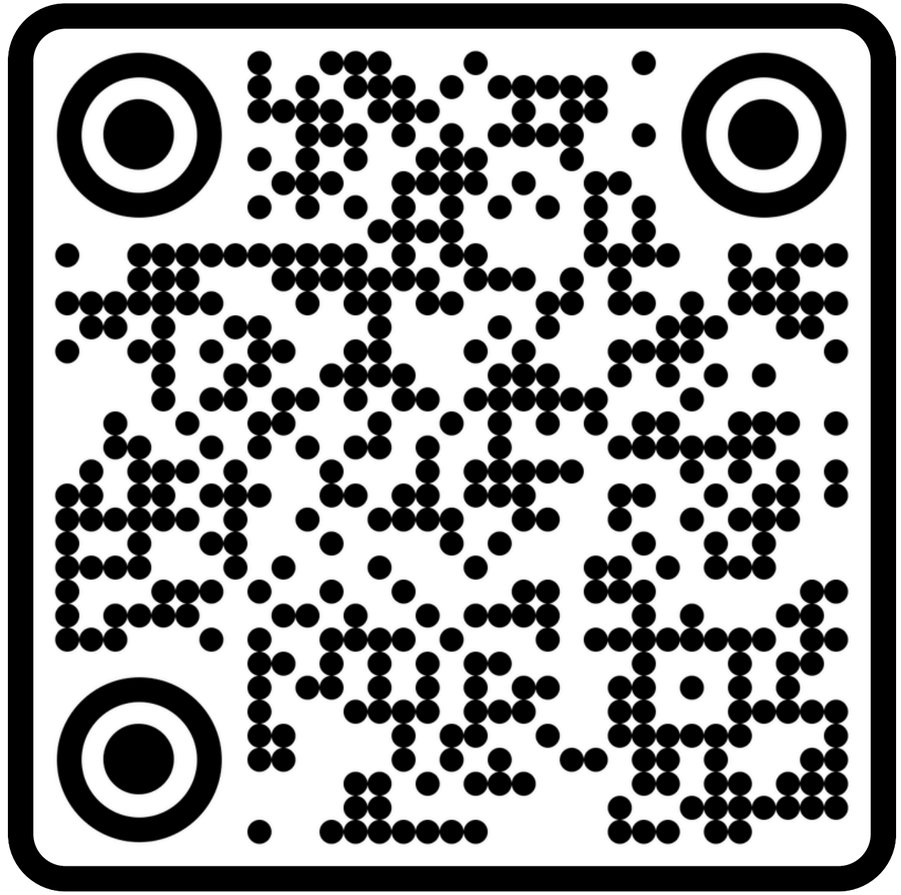
nf-core/mag: nf-core/mag is a reproducible pipeline for metagenome assembly, binning, and MAG analysis.

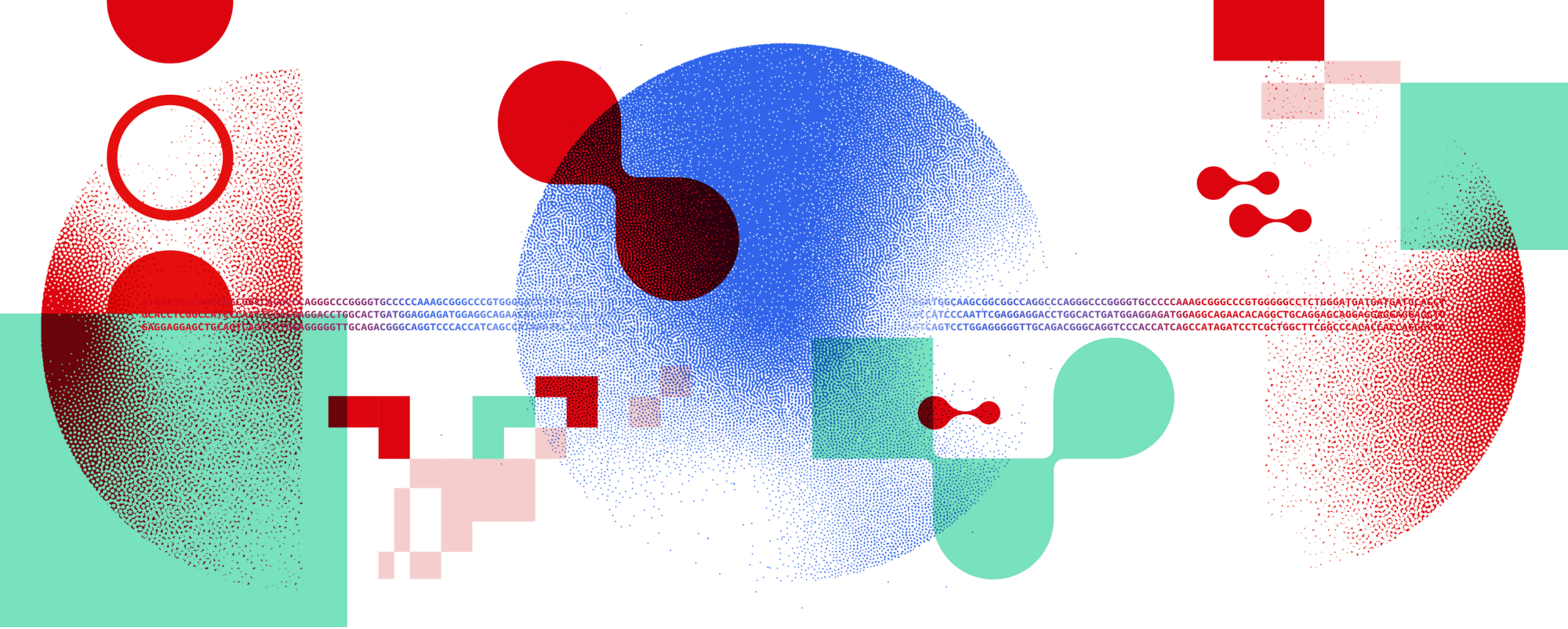


Summary

- Assembling and binning are the core steps for MAG recovery, and therefore the most computationally demanding.
- Selecting the correct pipeline requires careful analysis of multiples variables and requirements highly user-dependent.
- Tools/pipeline can have several parameters to fine-tune, try to test their effect on the final results do not trust them blindly.
- Try to use as many tools to measure MAG quality in order to assure confidence on your analysis: Time for Divide-and-Rule Tactics.

Practicals





Thank you

elixir SWITZERLAND

SIB